

Smart Air Purifier and AQI Monitoring System

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Abstract— Today, with growing numbers of people spending more of their times in the indoors, it is an major health concern. Low indoor air quality is another reason for respiratory diseases, heart diseases, and cancer cases. Air Purifier with AQI Monitoring - A smart air purifier and AQI monitoring system charged with removing pollutants from the air and displaying the air quality levels. Smart air purifiers leverage technology like UV-C lamps, activated carbon filters, and HEPA filters to rid airborne pollutants. HEPA filters trap minute air particles like dust, pollen and mould spores. These are enabled with activated carbon filters which can cold out odors, smoke from cigarettes, and volatile organic compounds etc. UV-C light removes dust, odors, and fungi, and also kills bacteria and viruses in the air. UV-A, and UV-B do not have this feature. Monitoring on AQI systems: AQI systems would have the ability to determine the concentration of air pollutants like carbonmonoxide, PM2. 10, and 2.5 and called them PM10, PM2.5, and PM1.5, respectively, at any single station, well as other pollutants in the atmosphere. Take advantage of this to determine where the pollution is coming from and address them specifically, to improve the air quality. For homes, businesses, schools, and other indoor spaces, there may be a place for smart air purifiers and AQI monitoring devices. People with allergies, asthma, and other respiratory issues really benefit from this. The size of the area that the system is suitable for, The types of pollutants you are (all up to you and the cost to the device).

Keywords- Air quality, wireless connections, device management, and the AQI system.

I. INTRODUCTION

Using the Air quality Index (AQI) along with some other Environmental hazards. As a result, it would need to decrease the amount of air pollutants quite significantly to minimize unhealthy air quality conditions [1]. This ultimately leads to significantly reduce the amount of deleterious compounds in the air in order to keep the air as healthy as it could get. Also, the data is stored using cloud computing as it is available anywhere [2]. When air pollution is detected, mobile applications aid in the information gathering process [3]. When air pollution is detected, mobile applications aid in the information gathering process. The EPA said in 1989 that certain pollutants may be present indoors at levels that are more than six times higher than they should be outdoors. Nonetheless, the U.S. EPA said in 1989 that indoor concentrations of certain toxins may be higher than outdoor concentrations. A home's features have a major impact on

both interior and outdoor pollution, according to Meieretal study of the PM10, PM2.5, and NO2 air quality inside and outside. The features of a house have a major impact on both interior and outdoor pollution, according to Meeretal study of PM10, PM2.5, and NO2 air quality inside and outside. When Giletal examined the effects of external pollution on interior air quality, he found that inside CO concentrations were greater than outside PM5 concentrations. Liuetal states that are dry throats, dry eyes & headache are the main three common symptoms that need to be The three most common symptoms are dry eyes, dry throats, and headaches, according to Liuetal. Indoor particles and both odorous and non- odorous gases were found in the air.

Carbon dioxide (CO2) and methane (CH4) CO-based sensors are essential for efficient real-time temperature and greenhouse gas monitoring. On the basis of this, Kashideta suggested an interior air quality system that monitored CO2, NO2, temperature, and dust. In addition, researchers have created a system for monitoring indoor air quality that combines wireless communication module, air quality sensors, and smart phone application. This system duplicates the data it collects and sends it to a cloud server for further analysis.

II. METHODS

A. Smart AQI Process:

The whole architecture system of the smart AQI system is shown below [12]. Indoor air quality monitoring system you described in your email is composed of a sensor, a wireless connection component, and an intelligent application component. The sensor part monitors the following: CO, CO2, PM10, PM2.5, PM1, humidity, temperature, and volatile organic compounds. The data measured is subsequently sent to the smart app via Bluetooth, Wi-Fi, or Lora, among other long- or short-distance communication methods. The user may see the measured data on the application's display. The programme gets notifications and hourly, daily, and yearly statistics from the server when it receives the measured data [15].

B. Smart Air Quality Monitoring System Hardware:

You specified in your email that the smart air quality apparatus consists of two components: the sensor and connectivity. The sensor component comprises a PM sensor (PM 2008M) designed to quantify PM 10 , PM 2.5 and PM 1, a CO2 carbon dioxide detection sensor employing non-

dispersive infrared absorption techniques to identify concentrations ranging from 0 to 200 ppm, and a carbon monoxide sensor with an electrochemical sensor capable of monitoring levels from 0 to 1000 ppm. Additionally, the sensor section incorporates Bluetooth and Wi-Fi modules to facilitate short-range communication, and a Lora module for long-range communication.

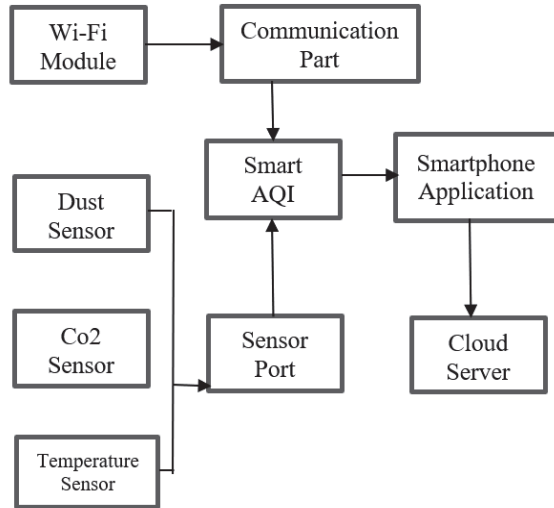


Figure 1: System Architecture of AQI

C. Smart AQI Monitor System Software:

The smart phone AQI's application collects data from the smart air quality monitor, shows it to the user clearly, and transfers it to a server. All users have access to the server's alerts, suggestions, and monitoring data. There are four displays in the air quality application: one for the weather, one for environmental data, and one for indoor and outdoor quality air. The gathered data by the specially designed smart air quality equipment is shown on the air quality indoor panel. This outside air quality panel displays the data from the Air Korea Open API. The Korea Meteorological Administration's open API is used to get weather data, which is shown on the weather screen [16]. The server operator provides and updates indoor environment data on the environmental information panel [10, 14].

D. Design and Working:

Via a front panel input, air enters the purifier and goes through a two-stage filtering procedure. The first step, known as the "Dust Filter," deals with impurities and big airborne particles that range in size from 3 to 10 mm. The "Pre Filter" then captures even smaller particles, including ones that are visible like hair, in the range of 10 to 25 microns. The great majority of particles bigger than three microns are captured by these first two filterstogether.

III. OUTCOMES AND DISCUSSION

The two parts of the smart air quality system you described in your email are the sensor and the communication. A PM sensor measures PM10, PM2.5, and PM1, a CO2 sensor uses non-dispersive infrared absorption to detect carbon dioxide at concentrations between 0 and 2000 ppm, a CO sensor uses electrochemistry to measure carbon monoxide levels between 0 and 1000 ppm, and a

VOCs sensor measures levels between a Bluetooth and Wi-Fi module for short-range communication and a Lora module for long-range communication are both included. The monitoring of pollutants such as PM2.5. When the PM 2.5 levels are elevated, or the purifier detects VOCs, it automatically adapts its filtration to best combat what it determines to be the most pressing threat to your indoor air quality. All of which adds up to you getting cleaner, healthier air to breathe. The AQI monitoring system detects real-time values of multiple parameters concerning Air Quality. With these, you can get to know how quality in the air changes in the indoors, to make decisions accordingly. Air Purifiers Smart Air Purifiers can turn their speed up or down automatically based on the live readings of AQI. Which means they use only as much power as it is actually need according to the air quality at any given moment, thus energy-saving. It can monitor how much you use the filters, as well as what the air quality is like, so it can tell you when you need to swap in a new one.

A. Smart Air Quality System Manufacturing:

This intelligent air quality monitor combines sensing and communication features into one compact device. Dust, temperature, humidity, and a variety of gases, such as carbon dioxide, carbon monoxide, volatile organic compounds (VOCs), and CO, may all be detected by its sensor array. Notably, the CM1106 carbon monoxide sensor measures CO concentrations between 0 and 2000 parts per million using non-dispersive infrared technology. It does this by using non-dispersive infrared absorption methods. The carbon monoxide sensor, an electrochemical device, monitors concentrations of the gas between 0 and 1000 parts per million. For the VOCs sensor, which uses electrochemical methods to track concentrations between 0 and 10 ppm, a semiconductor sensor is used. The digital sensor can detect temperatures between -40°C and 125°C and humidity levels between 0% and 100%. It is a temperature and humidity sensor. The communication section includes Bluetooth and Wi-Fi modules for short-range connections in addition to a Lora module for long-distance connectivity. When the preset standard levels are surpassed, red warnings appear to notify the user that the pollution limitations have been violated. The real-time measured data from the smart AQI equipment are then uploaded to the server via the smartphone app. Every piece of information is kept on file for any future projections.



Figure 2: Hardware for the smart AQI monitoring system.

B. AQI Fabrication:

Air quality's application fabrication is the process of designing and building tools and systems that may be used to assess and improve air quality. A variety of uses, such as indoor AQI monitors, might fall under this category. The monitors detect ground-level ozone, PM_{2.5}, PM₁₀, carbon monoxide, and volatile organic compounds (VOCs) in indoor air. The data they collect could lead to the identification of pollution blackspots and the introduction of targeted measures to clean the air, clever air cleaners. Air purifiers that are smart could be controlled through an app in a smartphone or operate on auto mode where they turn on and off which entails they will automatically react to the indoor air quality [6]. Approaches for air quality prediction Air quality management. These methods use present meteorological and air quality data to estimate future levels of air quality. This information can be used for people to plan activities around it and make measures to reduce their exposure to air pollution [7, 8]. Developing air quality applications can, therefore, be a daunting but necessary endeavour. We can improve the air quality.

C. AQI System Validation:

Validation of AQI system, Air quality Index VALIDATE an AQI system. This may be accomplished by comparing the system's AQI measurements to those of other air quality sensors or to known air quality standards. The verification of an AQI system could be done in a various way but here are some most popular methods :- Intercomparison:-It involves comparison of the AQI readings generated by the system with AQI readings from other air quality sensors which are believed to be more reliable [9]. This comparison can happen a million times at many times and places or all at once. Co-location refers to the practice of putting an AQI system near another, better quality reference air quality sensor. The reported AQI readings of the two monitors are matched against each other, after allowing for some time for the sensors to operate. Auditing: Review of data and documentation within the system to ensure it is being used properly and the information is being managed correctly [4]. Validating the AQI system is critical as it helps to ensure that the public receives valid and consistent air quality information. Both users who plan their activity and individuals suffering from health issues related to air quality, should now be aware of these values [5]. Validation of an AQI system Validate AQI System - Use a variety of validation strategies Validation is never perfect, so having as many ways as possible will give you a better picture of how accurate and reliable your system. Check the system often. Systems like the AQI ones keep getting updated and changed over time so one needs to make sure the system in question also returns the right information. Use different types of validation data. During the assessment, be sure to consider a wide-range and varied set of data - employing multiple different sensors for air quality and multiple instrumental locations. This works as a whole by doing it, so that you can make sure that the validation results will faithfully represent the features of the system.

IV. RESULT

The first column is the output of the smart air purifier, the second column is the proposed output of the smart air purifier, and the third column is the place of observation. The first row shows that the output of the smart air purifier

is 229 and the proposed output is 137. The place of observation is the classroom. The second row shows that the output of the smart air purifier is 236 and the proposed output is 230. The place of observation is the corridor. The third row shows that the output of the smart air purifier is 226 and the proposed output is 222. The place of observation is the workshop. The fourth row shows that the output of the smart air purifier is 560 and the proposed output is 560.

TABLE 1 : RESULT OF THE AQI

MQ – 135 Output	MQ – 135 Output(Proposed)	Place of Observation
229	137	Classroom
236	230 appr	Corridor
226	222 appr	Workshop
560(Polluted)	560(Polluted)	Roadside Area

V. CONCLUSION

Measuring PM₁₀, PM_{2.5}, carbon monoxide, and carbon dioxide was part of the dependability test for the "Amore" air quality system. Your stated sophisticated technology is designed to identify different types of indoor air pollution and notify users [11]. It keeps an eye on

humidity, temperature, and volatile organic compounds (VOCs) to provide a secure atmosphere. It makes use of both short-range (Bluetooth, Wi-Fi) and long-range (Lora) connections for the best possible connectivity. It may also be possible to forecast future air quality using this gathered data that is kept on a cloud server [13]. Analyzing the data transfer in detail and cross-referencing it with credible sources proved its correctness. This sophisticated system offers countermeasure suggestions along with monitoring data. Monitoring systems for air quality have the potential to both directly and indirectly lessen the negative effects of air pollution and greatly minimize indoor exposure. For two hours, data on temperature, humidity, carbon monoxide, and volatile organic compounds were gathered using Amore and other trustworthy measuring devices. Bramc Co., Ltd.'s BR-AIR-82K and Amore were used to measure PM₁₀ and PM_{2.5}. Carbon monoxide (CO₂) was detected using Amore from Testo Co. and the TSI-7525 from TSI Co., Ltd., while volatile organic compounds (VOCs) were quantified using Phoccheck Tiger and Amore from Senko Co., Ltd. Testo Co., Ltd.'s Amore and the Testo-608-H1 were both used to measure the temperature and humidity. The reliability differences between PM₁₀ and PM₂ Amore's temperature variances from other reliable devices were between 4% and 2% for humidity, 5% for CO, 1% for CO₂, 2% for volatile organic compounds, and so forth.

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